

# Observer-Patch Holography and the Dark Sector

## Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law

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August 20, 2026

**Paper release:** r2023   **Released:** August 20, 2026

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### Abstract

In observer-patch holography the Einstein equation sources curvature from the modular-charge stress of the screen under a universal coupling. Every screen source that carries modular charge therefore gravitates, whether or not it carries the electromagnetic readout, and the luminous-only Einstein relation holds exactly when the non-luminous charge vanishes. A dark sector is generic. The carrier is the anomalous modular energy on overlap collars, the same finite-stage remainder that the Einstein derivation controls: its rest-frame stress is bounded by the collar recovery defect, vanishes at full recovery, and dilutes as the inverse cube of the scale factor when conserved. The collars that carry it are galactic-scale objects, so a compact source sees only the ambient anomalous density; an isotropic ambient density has no quadrupole, and its tidal scale in the Solar System is four orders of magnitude below the Cassini bound. In the deep regime the anomalous density with linear enclosed mass gives the radial-acceleration relation  $a = \sqrt{a_b a_0}$ , flat rotation curves, and the baryonic Tully–Fisher relation  $v^4 = GM_b a_0$ ; on the SPARC sample both relations are reproduced with one constant near  $1.4 \times 10^{-10} \text{ m s}^{-2}$ . All structural statements are machine-checked. The magnitude of the anomalous charge, its full radial profile, and the collar-scale theorem are work in progress.

## Claim Boundary

The finite OPH results used here are the rest-frame Einstein relation on its named small-ball premises, the decomposition of the cap modular generator into a bulk part, a central part, and a collar-carried anomalous part, and the controlled bound on that anomalous part [1, 2]. Every theorem and proposition below is an algebraic or analytic consequence of those objects together with the premises named in the text, and each is checked in the Lean formalization [3]. Universal coupling, the collar-scale premise, and the linear enclosed-mass premise are structure fields or named hypotheses in that formalization; none is established there.

The paper does not supply a quotient-invariant receipt localizing the anomalous modular energy on collars, the value of the collar constant, the profile of the anomalous density outside the deep regime, a proof that the collars carrying the anomaly are galactic-scale, or a cosmological likelihood. The acceleration constant  $a_0$  is a fitted comparison value. No cosmological result carries an OPH falsification verdict.

# Contents

|           |                                                    |          |
|-----------|----------------------------------------------------|----------|
| <b>1</b>  | <b>What Sources Gravity in OPH</b>                 | <b>2</b> |
| <b>2</b>  | <b>The Generic Dark Sector</b>                     | <b>2</b> |
| <b>3</b>  | <b>The Modular-Anomaly Source</b>                  | <b>3</b> |
| <b>4</b>  | <b>Galactic-Scale Collars and the Solar System</b> | <b>4</b> |
| <b>5</b>  | <b>The Deep-Regime Profile</b>                     | <b>4</b> |
| <b>6</b>  | <b>Lensing and the Relativistic Source</b>         | <b>5</b> |
| <b>7</b>  | <b>Comparison with Rotation-Curve Data</b>         | <b>6</b> |
| <b>8</b>  | <b>Cosmological Scaling</b>                        | <b>6</b> |
| <b>9</b>  | <b>Open Targets</b>                                | <b>6</b> |
| <b>10</b> | <b>Conclusion</b>                                  | <b>7</b> |

## 1 What Sources Gravity in OPH

The Einstein arrow of the spacetime paper ends in

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \quad (1)$$

where  $\langle T_{ab} \rangle$  is the local conserved stress reconstructed from positive null translations and modular charges of the record algebra [1]. In the rest frame the relation follows from three premises on a small geodesic ball of radius  $\ell$ : generalized entropy is stationary, the bulk entropy variation equals  $(8\pi^2 \ell^4/15) \delta \langle T_{00} \rangle$ , and the fixed-volume area variation equals  $-(4\pi \ell^4/15) \delta G_{00}$ . Exact arithmetic gives  $\delta G_{00} = 8\pi G \delta \langle T_{00} \rangle$ .

The stress entering this relation is modular charge. The screen does not consult the electromagnetic readout of a source before responding to it. That is the universal-coupling premise, and it is the only input the dark-sector question takes from the gravity derivation.

## 2 The Generic Dark Sector

Let a finite family of screen sources be indexed by  $i$ , each with rest-frame modular-charge contribution  $q_i$  and a flag recording whether it carries the electromagnetic readout. Write  $Q_{\text{lum}}$  for the sum over flagged sources and  $Q_{\text{dark}}$  for the sum over the others, so that  $Q_{\text{tot}} = Q_{\text{lum}} + Q_{\text{dark}}$ .

**Premise 2.1** (Universal coupling). *The stress contraction entering the rest-frame Einstein relation equals  $Q_{\text{tot}}$ , with no readout-dependent weight.*

**Theorem 2.2** (Geometry responds to all modular charge). *Under Premise 2.1,  $\delta G_{00} = 8\pi G (Q_{\text{lum}} + Q_{\text{dark}})$ .*

**Theorem 2.3** (The luminous-only relation). *Under Premise 2.1,  $\delta G_{00} = 8\pi G Q_{\text{lum}}$  holds if and only if  $Q_{\text{dark}} = 0$ .*

**Corollary 2.4** (A geometric excess forces a dark source). *If  $\delta G_{00} \neq 8\pi G Q_{\text{lum}}$ , some source without the electromagnetic readout carries nonzero modular charge.*

**Proposition 2.5** (Attractive sign). *If every non-luminous charge is nonnegative,  $Q_{\text{dark}} \geq 0$ , and the dark contribution enters the weak-field equation with the sign of baryonic mass.*

*Proofs.*  $Q_{\text{tot}} = Q_{\text{lum}} + Q_{\text{dark}}$  is a finite-sum identity. Theorem 2.2 substitutes it into the rest-frame relation. Theorem 2.3 cancels  $8\pi G > 0$ . The corollary is the contrapositive of the sum of zeros, and the proposition is a sum of nonnegative terms.  $\square$

In a theory where gravity is sourced by modular charge, the existence of a dark sector is settled by these four lines: any geometric excess over the luminous prediction is by itself the presence of non-luminous modular charge. The remaining questions are which screen object carries that charge, how much of it there is, and how it is distributed.

### 3 The Modular-Anomaly Source

On the support-visible branch the cap modular generator has the local form

$$K_C = 2\pi B_C + K_C^{\text{anom}} + \text{const}, \quad (2)$$

with  $B_C$  the bulk boost generator and  $K_C^{\text{anom}}$  the nonadditive part carried on the overlap collar [1]. At finite regulator stage the Einstein derivation carries this part as a remainder and bounds it:

$$|\delta \langle E_C^{(\delta)} \rangle| \leq C_C \eta_\delta, \quad (3)$$

where  $\eta_\delta$  is the collar recovery defect and  $C_C$  depends only on the collar model and the bounded observable class.

**Definition 3.1** (Anomalous stress and density).

$$\delta \langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2 \ell^4} \delta \langle E_C^{(\delta)} \rangle, \quad \rho_A := \frac{1}{c^2} \delta \langle T_{00}^{\text{anom}} \rangle. \quad (4)$$

The dark-sector identification is that this anomalous modular energy is the non-luminous charge of Section 2. It carries no electromagnetic readout by construction: it is the part of the modular generator that no local record exposes. Three consequences follow from (3) and (4).

**Theorem 3.2** (Bounded source).  $|\delta \langle T_{00}^{\text{anom}} \rangle| \leq 15 C_C \eta_\delta / (8\pi^2 \ell^4)$ .

**Theorem 3.3** (Full recovery switches the source off). *If  $\eta_\delta = 0$  then  $\delta \langle T_{00}^{\text{anom}} \rangle = 0$ . Along any family with fixed  $\ell$ , bounded  $C_C$ , and  $\eta_\delta \rightarrow 0$ , the anomalous stress tends to zero.*

**Theorem 3.4** (Cold scaling). *A conserved anomalous charge in a comoving cell of physical volume  $\propto a^3$  has density  $\propto a^{-3}$ .*

*Proofs.* Theorem 3.2 multiplies (3) by the positive kernel inverse. Theorem 3.3 is the case  $\eta_\delta = 0$  of the bound and a squeeze along the family. Theorem 3.4 is  $\rho a^3 = Q$ .  $\square$

In the weak-field limit the anomalous density enters the Poisson equation beside the baryons,

$$\nabla^2 \Phi = 4\pi G (\rho_b + \rho_A), \quad (5)$$

with the attractive sign of Proposition 2.5 whenever the anomalous charge is nonnegative.

## 4 Galactic-Scale Collars and the Solar System

The collars in (2) are overlap collars between observer patches, and the recovery defect  $\eta_\delta$  measures how far the repaired record on a collar falls short of determining its complement. Where records are dense and repair saturates, the defect is small; where the screen is sparsely recorded, over the settled outskirts of a galaxy, the defect is carried on long-lived cuts. The anomalous charge is therefore a property of the record state of a region, and the region in question is the galaxy.

**Premise 4.1** (Collar scale). *The collars carrying  $\delta \langle E_C^{(\delta)} \rangle$  are galactic-scale settled cuts. A compact source inside the galaxy does not generate anomalous charge of its own; it sits in the ambient anomalous density  $\rho_A$  of its neighbourhood.*

Under this premise the Solar System is a test of the ambient density alone.

**Proposition 4.2** (An isotropic ambient density has no quadrupole). *The potential of a locally uniform density is  $(2\pi G \rho_A / 3) r^2$ . Its projection on the quadrupole weight  $P_2(\cos \theta)$  vanishes,  $\int_{-1}^1 (\mu^2 - \frac{1}{3}) d\mu = 0$ , so the quadrupole coefficient  $Q_2$  is zero. The only footprint is the isotropic tidal scale  $4\pi G \rho_A / 3$ .*

With the local dark density  $\rho_A \simeq 6.8 \times 10^{-22} \text{ kg m}^{-3}$ , the value inferred from Galactic kinematics [8], the tidal scale is

$$\frac{4\pi G \rho_A}{3} = 1.9 \times 10^{-31} \text{ s}^{-2}, \quad (6)$$

against the Cassini quadrupole bound  $Q_2 = (1.6 \pm 1.8) \times 10^{-27} \text{ s}^{-2}$  [7]: four orders of magnitude below the uncertainty, with zero quadrupole. The anomalous mass enclosed within ten astronomical units is  $5 \times 10^{-15} M_\odot$ . Both numbers are machine-checked inequalities in the formalization. The Solar System constrains the premise of Section 4 and nothing else in the construction.

## 5 The Deep-Regime Profile

Around a baryonic mass  $M_b$ , the codimension-one collar count of the microphysics paper [2] gives the number of settled cuts enclosing the source within radius  $r$  as proportional to  $r$ , each cut carrying a fixed anomalous charge set by  $M_b$  and the acceleration constant.

**Premise 5.1** (Linear enclosed mass). *In the deep regime the anomalous mass enclosed within radius  $r$  is*

$$M_A(r) = r\sqrt{\frac{M_b a_0}{G}}, \quad \rho_A(r) = \frac{1}{4\pi r^2} \sqrt{\frac{M_b a_0}{G}}. \quad (7)$$

The density is the shell derivative of the enclosed mass,  $M'_A(r) = 4\pi r^2 \rho_A(r)$ .

**Theorem 5.2** (Deep radial-acceleration relation). *The anomalous acceleration of the profile (7) is*

$$a_A(r) = \frac{GM_A(r)}{r^2} = \frac{\sqrt{GM_b a_0}}{r} = \sqrt{a_b(r) a_0}, \quad a_b = \frac{GM_b}{r^2}. \quad (8)$$

**Theorem 5.3** (Flat rotation curve and baryonic Tully–Fisher relation). *The circular speed supported by  $a_A$  satisfies  $v^2 = a_A r = \sqrt{GM_b a_0}$ , independent of  $r$ , and hence*

$$v^4 = GM_b a_0. \quad (9)$$

*Proofs.*  $GM_A/r^2 = G\sqrt{M_b a_0}/G/r = \sqrt{GM_b a_0}/r$ , and  $\sqrt{GM_b a_0}/r = \sqrt{(GM_b/r^2)a_0}$  for  $r > 0$ . Multiplying by  $r$  removes the radius; squaring gives (9).  $\square$

The exponents 1/2 in (8) and 4 in (9) are the parameter-free content of the profile. The constant  $a_0$  collects  $C_C$ , the collar density, and the small-ball normalization; none of these is fixed by the source at present, so  $a_0$  enters every comparison as a fitted value. The deep regime is the range  $a_b \ll a_0$ , equivalently  $r \gg r_M = \sqrt{GM_b/a_0}$ , where the anomalous acceleration dominates the baryonic one. Inside  $r_M$  the enclosed anomalous mass is small compared with  $M_b$  and the response is Newtonian; the profile between the two regimes is not fixed by Premise 5.1.

## 6 Lensing and the Relativistic Source

The anomalous charge enters (1) through the same  $\langle T_{ab} \rangle$  that sources the timelike response. Under Premise 2.1 there is one stress tensor, and the null directions of the spacetime paper’s tomography frame read the same charge as the timelike ones [1].

**Proposition 6.1** (One source for lensing and dynamics). *Under Premise 2.1, the mass inferred from null geodesics through a region equals the mass inferred from timelike orbits in it: both are  $Q_{\text{lum}} + Q_{\text{dark}}$ .*

*Proof.* Both readouts are contractions of the one tensor in (1) with the respective tangent vectors, and the premise supplies no readout-dependent weight.  $\square$

The proposition is structural. The relativistic refinement limit that turns the weak-field density (7) into a full  $T_{ab}^{\text{anom}}$ , with its pressure and anisotropic stress, is work in progress; until then the lensing statement is that the anomalous charge lenses with the same weight with which it moves stars.

## 7 Comparison with Rotation-Curve Data

Theorems 5.2 and 5.3 are compared with the SPARC sample of 175 disk galaxies with  $3.6\mu\text{m}$  photometry [5] under the standard cuts of McGaugh, Lelli, and Schombert [6]: quality flag at most 2, inclination at least  $30^\circ$ , velocity error at most ten percent, stellar mass-to-light ratios 0.5 for disks and 0.7 for bulges, leaving 2700 points in 149 galaxies. The computation is released with the paper [4].

**Radial acceleration.** On the self-consistent deep subset  $g_{\text{bar}} < f a_0$ , the fixed-exponent fit  $g_{\text{obs}} = \sqrt{g_{\text{bar}} a_0}$  gives

$$a_0 = 1.57 \times 10^{-10}, \quad 1.40 \times 10^{-10}, \quad 1.28 \times 10^{-10} \text{ m s}^{-2}$$

for  $f = 0.3, 0.1$ , and  $0.03$ , with 1651, 1043, and 233 points and rms scatter between 0.14 and 0.17 dex. The free exponents on the same subsets are  $0.60 \pm 0.01$ ,  $0.57 \pm 0.02$ , and  $0.39 \pm 0.08$ , with point-bootstrap errors. The exponent moves toward  $1/2$  as the cut deepens and the sample runs out of points before the deep regime is clean; the data are consistent with exponent  $1/2$  at the available depth, with the constant between  $1.3$  and  $1.6 \times 10^{-10} \text{ m s}^{-2}$  depending on the cut.

**Baryonic Tully–Fisher.** For the 123 galaxies with a measured flat velocity that pass the same cuts, with  $M_b = 0.5L_{[3.6]} + 1.33M_{\text{HI}}$ , the fixed-exponent normalization  $a_0 = v_f^4/(GM_b)$  gives  $a_0 = 1.55 \times 10^{-10} \text{ m s}^{-2}$  with 0.26 dex scatter, and the free exponent from an ordinary least-squares fit of  $\log v_f$  on  $\log M_b$  is  $3.75 \pm 0.10$ .

**One constant.** Theorems 5.2 and 5.3 share one  $a_0$ . The constants recovered from the two relations differ by 0.046 dex at  $f = 0.1$ , inside the combined scatter. This is the one nontrivial check the deep profile admits with a single fitted constant, and it passes.

## 8 Cosmological Scaling

If the total anomalous charge in a comoving region is conserved, which holds whenever the collar family of the region is stable under the expansion, Theorem 3.4 gives  $\rho_A \propto a^{-3}$ : the anomalous medium is pressureless on cosmological scales. With Premise 4.1 it clusters with the galaxies whose collars carry it. The abundance  $\Omega_A$ , the perturbation initial data, and the transfer functions require the collar density of the early screen and are work in progress; the scaling is the statement that the medium is cold.

## 9 Open Targets

1. **Collar scale.** Prove Premise 4.1: that the recovery defect  $\eta_\delta$  saturates to zero on the collars of a compact source and is carried on galactic-scale settled cuts, with a rate in the local record density. Theorem 3.3 then makes the Solar-System footprint of Section 4 a theorem rather than a premise.

2. **Magnitude.** Attach the normalization of  $\delta\langle E_C^{(\delta)}\rangle$  to repair conservation on the Ward/Bianchi rung of the gravity premise ladder, so that  $a_0$  becomes a consequence of  $C_C$  and the collar density.
3. **Profile.** Derive the linear enclosed mass of Premise 5.1 from the collar geometry of a point source, and extend it through the transition to the Newtonian regime.
4. **Localization.** Supply the quotient-invariant receipt identifying the collar operator in (2) with the small-ball remainder in (4).
5. **Abundance.** Fix the comoving anomalous charge of the early screen and the resulting  $\Omega_A$ , then the joint galaxy, cluster, lensing, CMB, and growth likelihood.

## 10 Conclusion

Under universal coupling the screen gravitates according to its total modular charge, so a dark sector is a theorem-level feature of OPH: any geometric excess over the luminous prediction is the presence of non-luminous modular charge. The anomalous modular energy on overlap collars is the carrier the Einstein derivation itself supplies. It is bounded by the recovery defect, absent at full recovery, cold when conserved, and, on galactic-scale collars, invisible to the Solar System by four orders of magnitude. Its deep-regime profile gives the radial-acceleration relation, flat rotation curves, and the baryonic Tully–Fisher relation with one constant, and the SPARC sample returns that constant consistently from both relations. The collar-scale theorem, the magnitude of the charge, and the full profile are the work that turns the premises of Sections 4 and 5 into derivations.

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## AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic’s Fable and OpenAI’s GPT-5.6-Sol, for research support, software development, editing, and synthesis. The authors are responsible for the paper’s claims, methods, and final text.